

```

//////////////////////////////////// MAIN
MODULE////////////////////////////////////
`timescale 1ns / 1ps
////////////////////////////////////
////
// Company:
// Engineer:
//
// Create Date: 05.10.2026 14:45:26
// Design Name:
// Module Name: traffic_light_controller
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
////////////////////////////////////
////

module traffic_light_controller (
    input clk,
    input reset,
    input pedestrian,
    output reg [2:0] highway,
    output reg [2:0] side_road,
    output reg ped_walk
);
//-----
// Light Encoding
//-----
parameter RED = 3'b100;
parameter YELLOW = 3'b010;
parameter GREEN = 3'b001;
//-----
// FSM States
//-----
parameter HW_GREEN = 2'd0;
parameter HW_YELLOW = 2'd1;
parameter SR_GREEN = 2'd2;
parameter PED_STATE = 2'd3;
reg [1:0] state;
reg [3:0] timer;
//-----
// FSM
//-----
always @(posedge clk or posedge reset)
begin
    if(reset)
    begin
        state

```